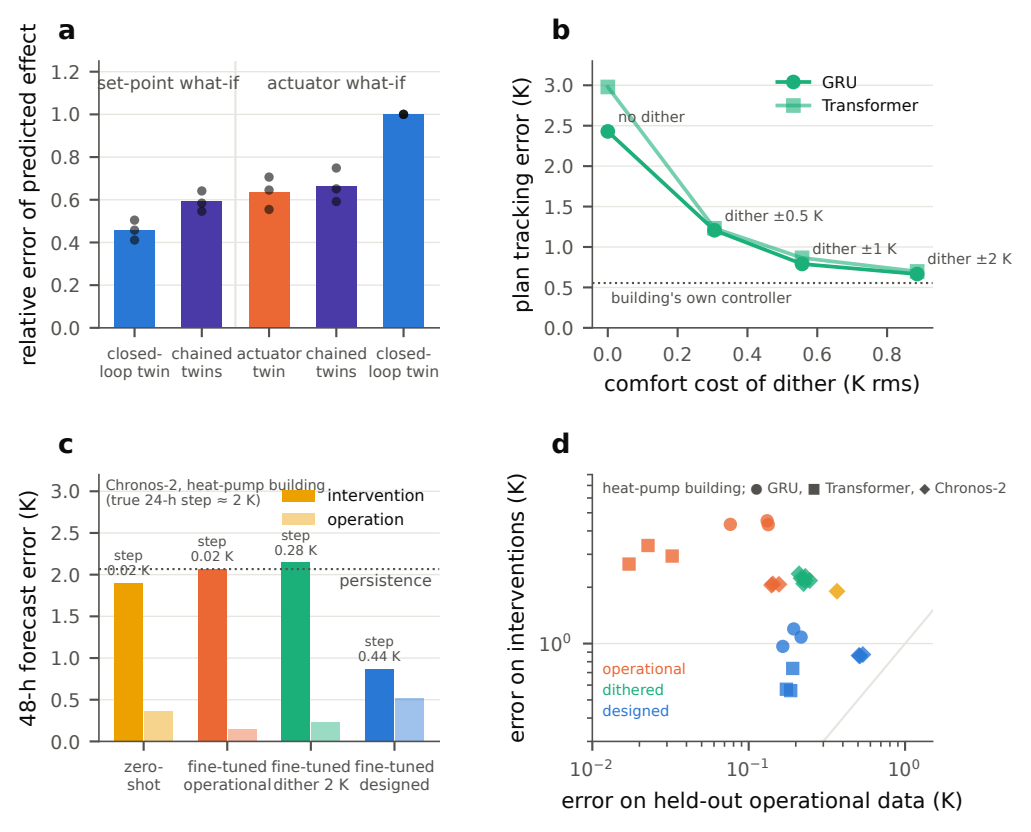
a
relative error of predicted effect
1.2
1.0
0.8
0.6
0.4
0.2
0.0
set-point what-if
actuator what-if
closed-loop twin
chained twins
actuator twin
chained twins
closed-loop twin
b
plan tracking error (K)
3.0
2.5
2.0
1.5
1.0
0.5
0.0
GRU
Transformer
no dither
dither ±0.5 K
dither ±1 K
dither ±2 K
building's own controller
0.0
0.2
0.4
0.6
0.8
comfort cost of dither (K rms)
c
48-h forecast error (K)
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Chronos-2, heat-pump building
(true 24-h step ≈ 2 K)
intervention
operation
step 0.02 K
step 0.02 K
step 0.28 K
step 0.44 K
persistence
zero-shot
fine-tuned operational
fine-tuned dither 2 K
fine-tuned designed
d
error on interventions (K)
heat-pump building; ● GRU, ■ Transformer, ◆ Chronos-2
10^0
operational
dithered
designed
10^{-2}
10^{-1}
10^0
error on held-out operational data (K)